

Global Horizontal Irradiance

Global Horizontal Irradiance (GHI) is the amount of terrestrial irradiance falling on a surface horizontal to the surface of the earth. GHI can be measured with a variety of instruments. The most common instrument used to measure GHI is called a pyranometer which has a hemispherical (180°) view angle. The hallmark of a pyranometer is a true cosine response to incident angle, i.e. the response of the pyranometer to a beam of light is proportional to the cosine of the incident angle of the beam. Most pyranometers utilize a thermopile sensor to sense incoming light; however, pyranometers such as the Licor LI-200 use a photovoltaic device with a diffuser. Thermopile-based pyranometers have a flat spectral response to incoming light, while PV-based pyranometers will have spectral sensitivity according to the PV materials used in the sensor. Pyranometers using a PV sensor respond nearly instantaneously to changes in irradiance, unlike their thermopile counterparts which take anywhere from 1 second to 15 seconds to achieve full response to a step change in irradiance. GHI may also be measured with a photovoltaic reference cell, which will have spectral sensitivity and generally will not exhibit true cosine response.

If GHI cannot be measured directly, it may be calculated from direct normal irradiance (DNI) and diffuse horizontal irradiance (DHI) using the following equation:

$$GHI = DHI + DNI \cdot \cos(\theta_Z)$$

In some PV systems, GHI is measured and a model, such as the DISC or [DIRINT](#) models, is used to estimate the DNI or DHI.

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